Making Luminis Glow Mods by Hand

This guide shows how to make a Luminis glow mod using only **Penumbra** and an **image editor**: no scripts, no command line. The main walkthrough makes a glowing tattoo; the variations after it cover eyes, hair, and gear, minions and mounts, which work the same way.

What you need

- The **Luminis** mod installed and enabled
- Penumbra (and the mod your glow builds on, for example TBSE or Bibo for tattoos)
- An image editor that exports PNG with transparency: Photoshop, GIMP, Krita or Paint.NET

How a glow mod works

A glow mod contains just two files:

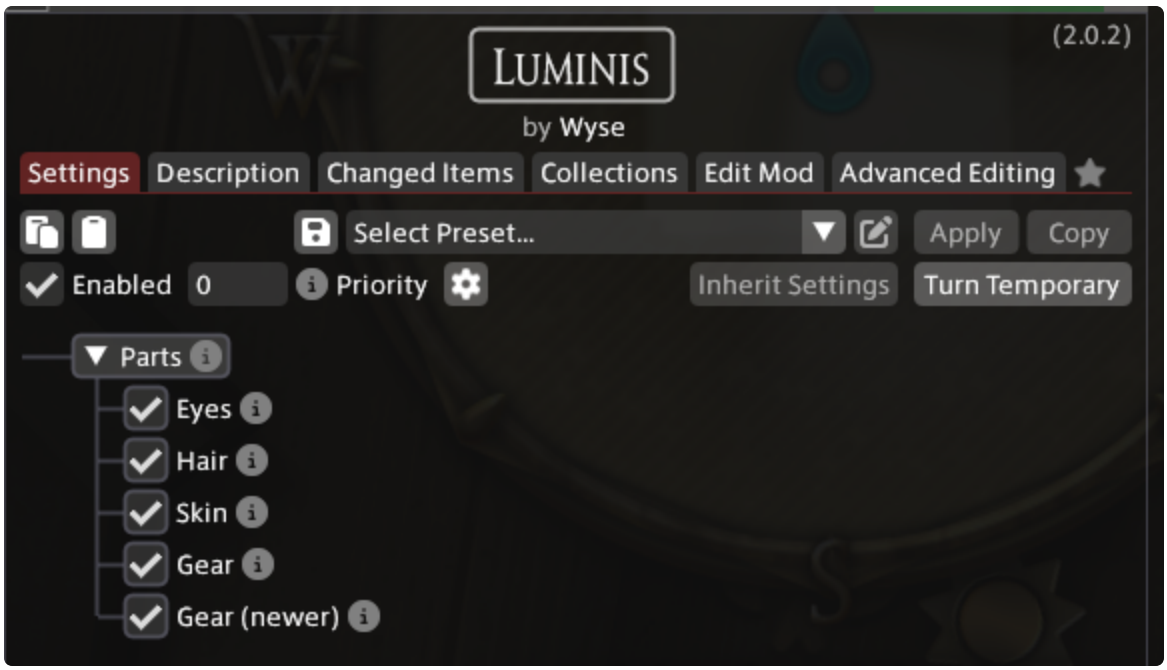
1. **A copy of the material** your glow goes on (the body's skin material for a tattoo), with Luminis settings turned on.
2. **A glow layer texture**: an image where the glowing parts are painted in the colors they should glow, and everything else is transparent.

The material points at the glow layer, and its Luminis settings decide how bright the glow is and when it appears. Because the glow lives in its own mod, it can be switched on and off without touching the tattoo or body mod.

Before you start: install Luminis

1. Import the Luminis `.pmp` in Penumbra (drag it onto the mod list) and enable it.
2. In its **Settings**, under **Parts**, keep on what you'll use: **Eyes**, **Hair**, **Skin**, **Gear** (most gear, minions and mounts) and **Gear (newer)**.
3. **Restart the game.** Shaders only load when the game starts, so do this every time you install or update Luminis. Until you restart, Penumbra already shows the new settings, but the game still draws with the old ones.
4. Check it's running: in the **On-Screen** tab, open your eyes or body, and the **Shader Package** row should say **[Luminis]**.

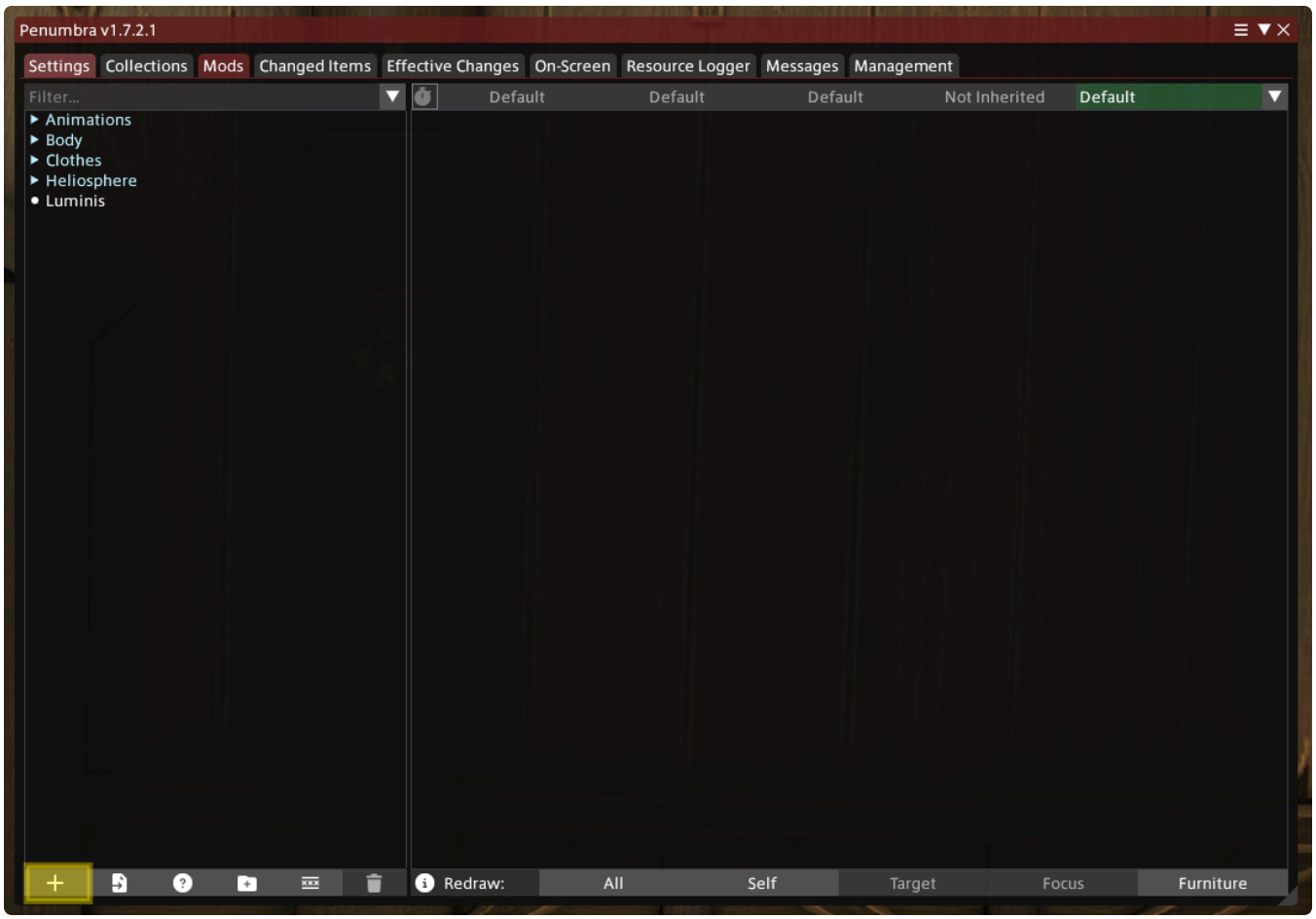
Keep only one copy of Luminis enabled. Installing it changes nothing on its own: nothing glows until you set it up, as below.



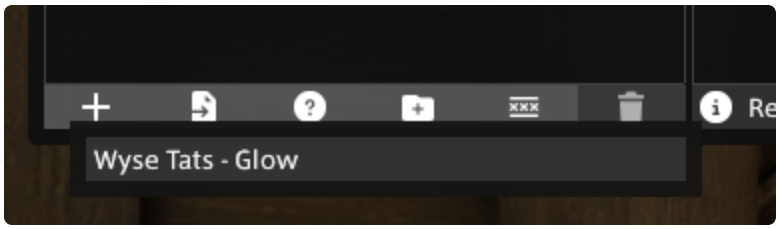
Step 1: Create the mod and copy in the material

Before you start, put on the body mod and tattoo your glow is for, and **disable any existing glow mod for them**, so you copy the body mod's original material. Type `/penumbra redraw self` after changing mods.

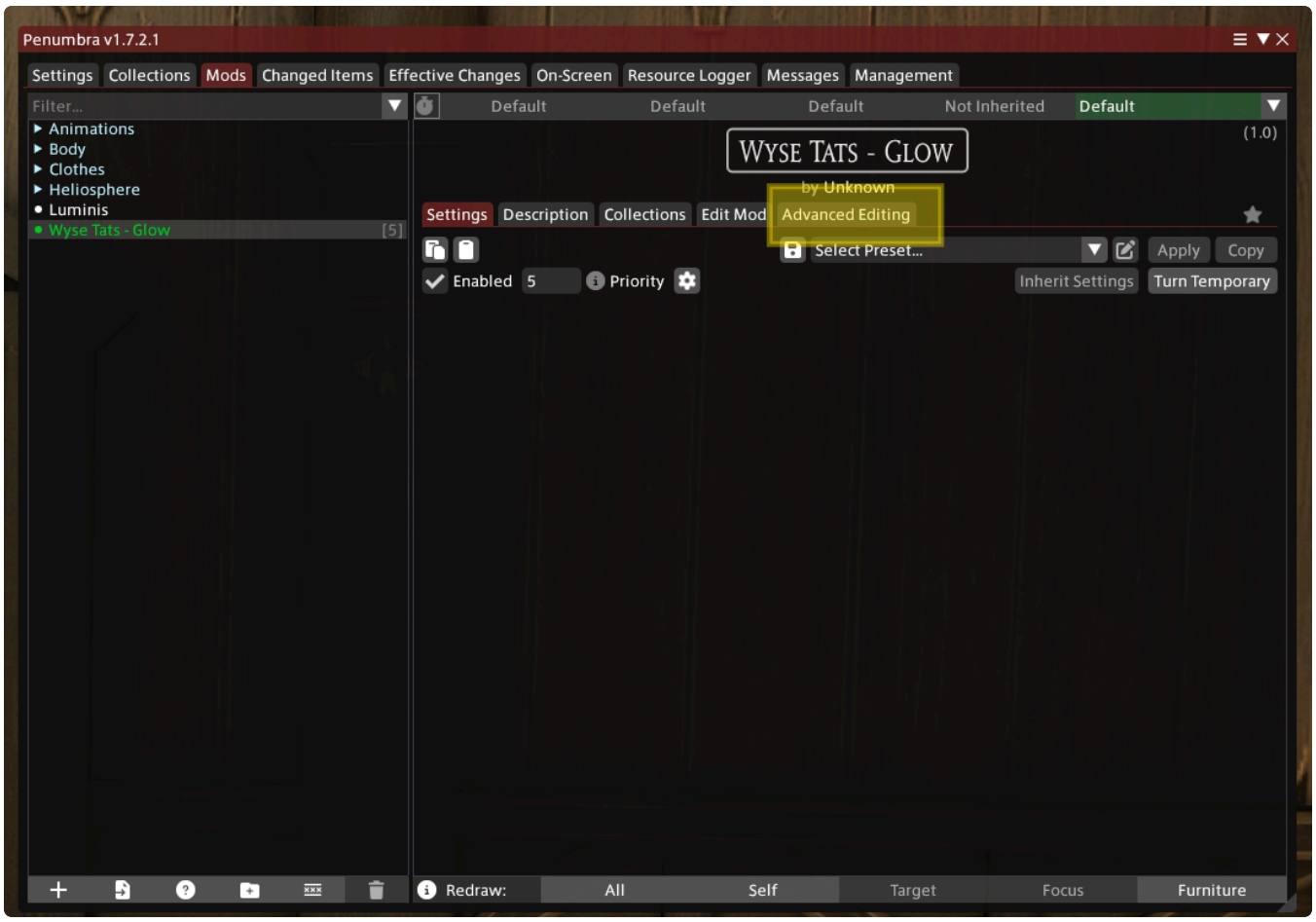
1. In Penumbra's **Mods** tab, click the **+** button at the bottom of the mod list.



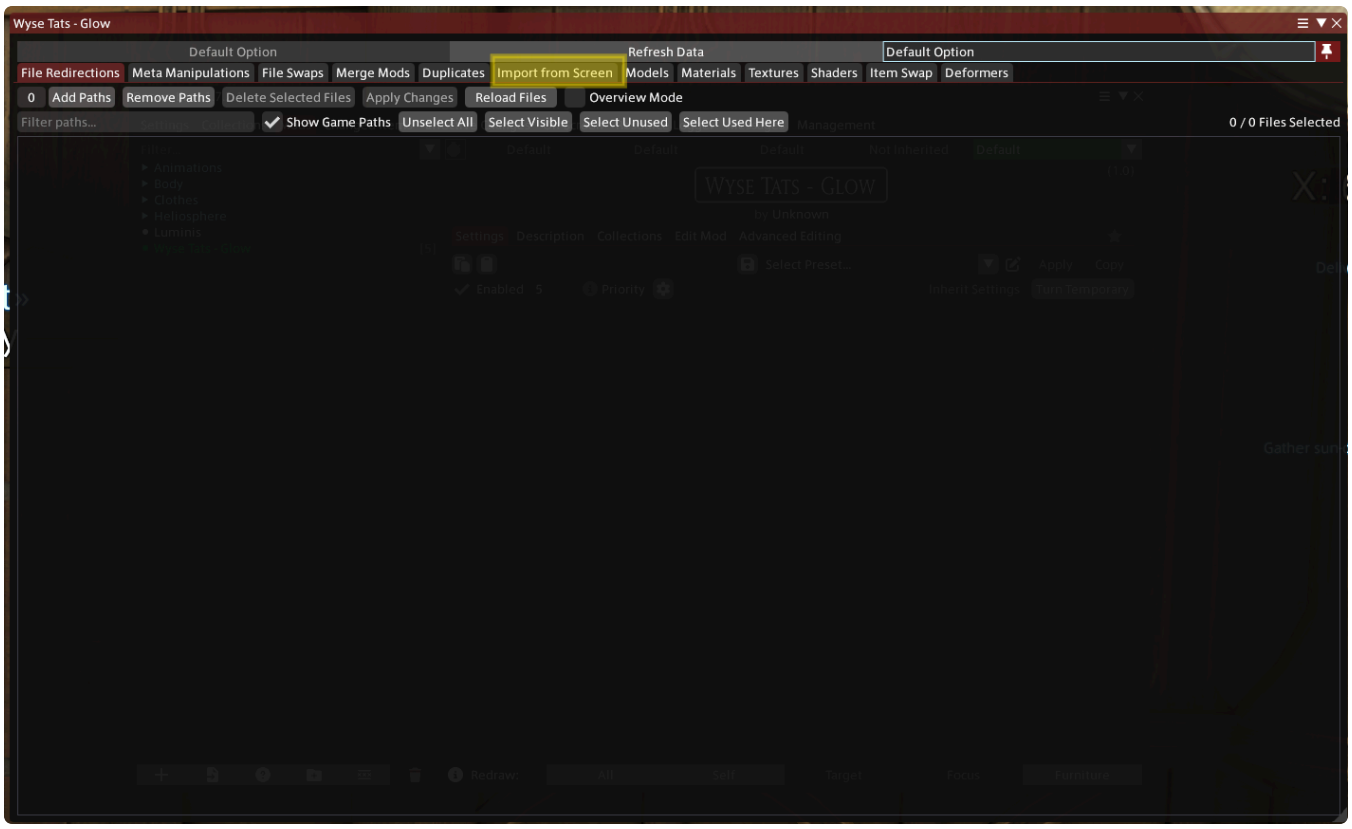
2. Type a name for the new mod, for example `My Tattoo - Glow`, and press **Enter**.



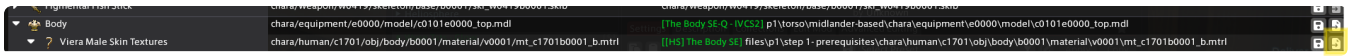
3. Select the new mod and open its **Advanced Editing** tab.



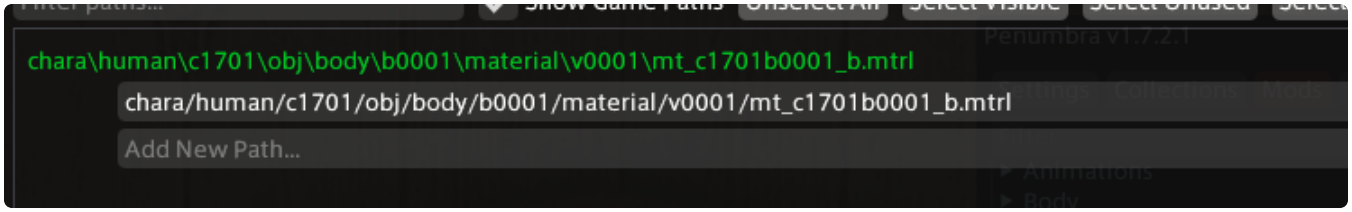
4. In the editing window, open the **Import from Screen** tab. It lists everything your character is currently using.



5. Expand **Body**, and find the **Skin Textures** row with the body's skin material (named like `mt_c1701b0001_b.mtrl`). The right-hand column shows which mod it comes from; it should be your body mod, for example **[HS] The Body SE**. Click the **import button** at the far right of that row (next to the save icon).



6. Open the **File Redirections** tab. The material is now part of your mod, listed in green with its game path.

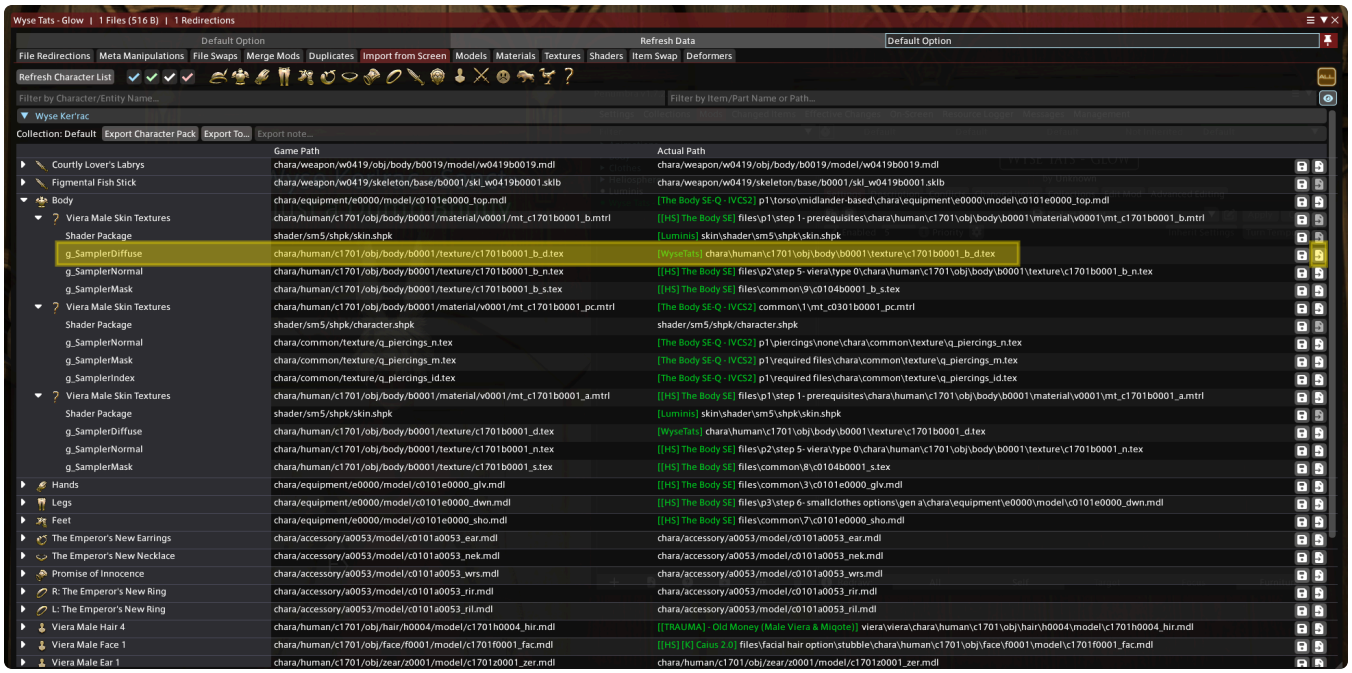


Body mods often split the body across several materials. Pick the one that covers the tattoo: for TBSE, that's the `_b` material. Not sure? Check the **On-Screen** tab, where each **Skin Textures** entry is one material, with its textures listed underneath.

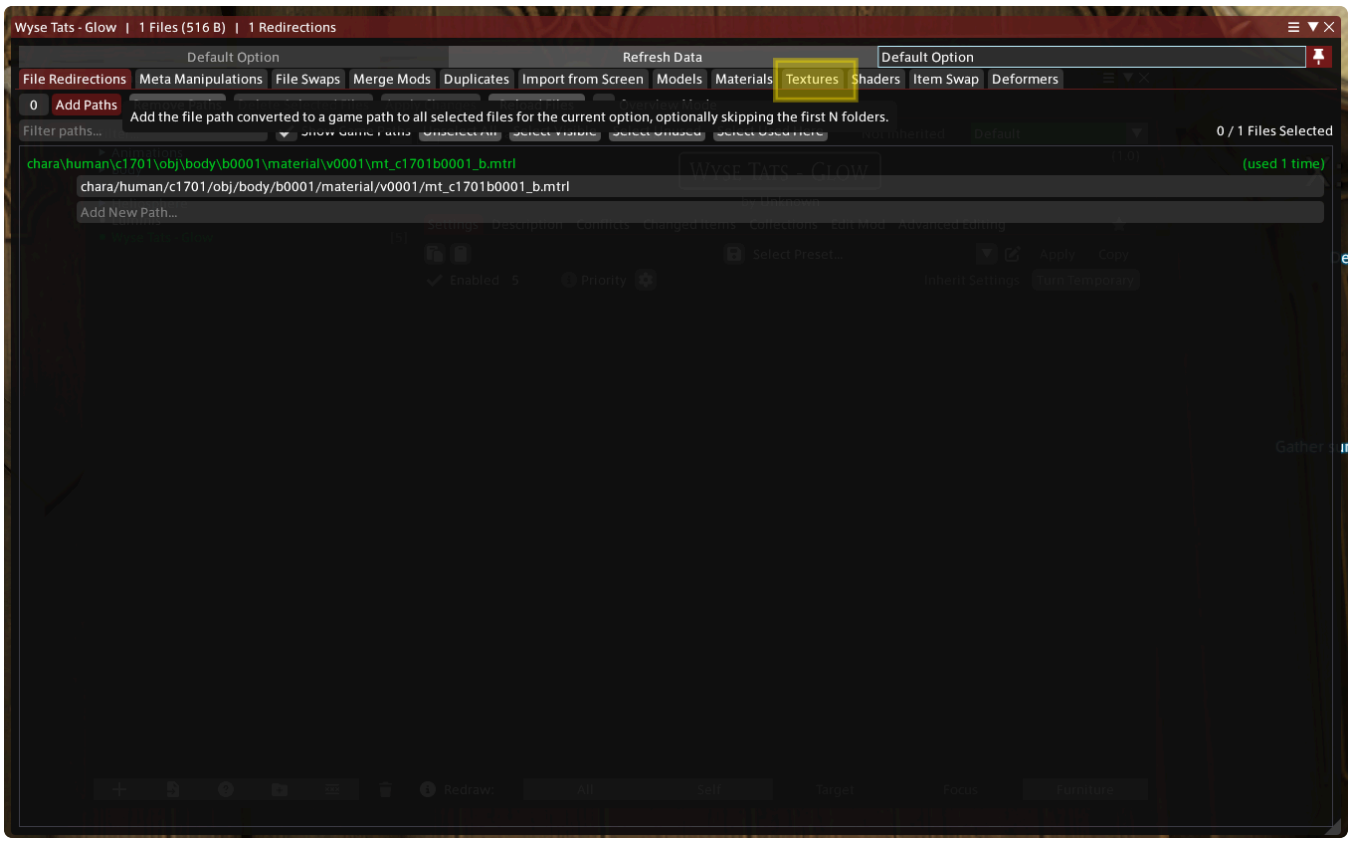
Step 2: Paint the glow layer

First, get a layout reference: the color texture your character is wearing right now, tattoo included, so you can paint the glow right over its lines.

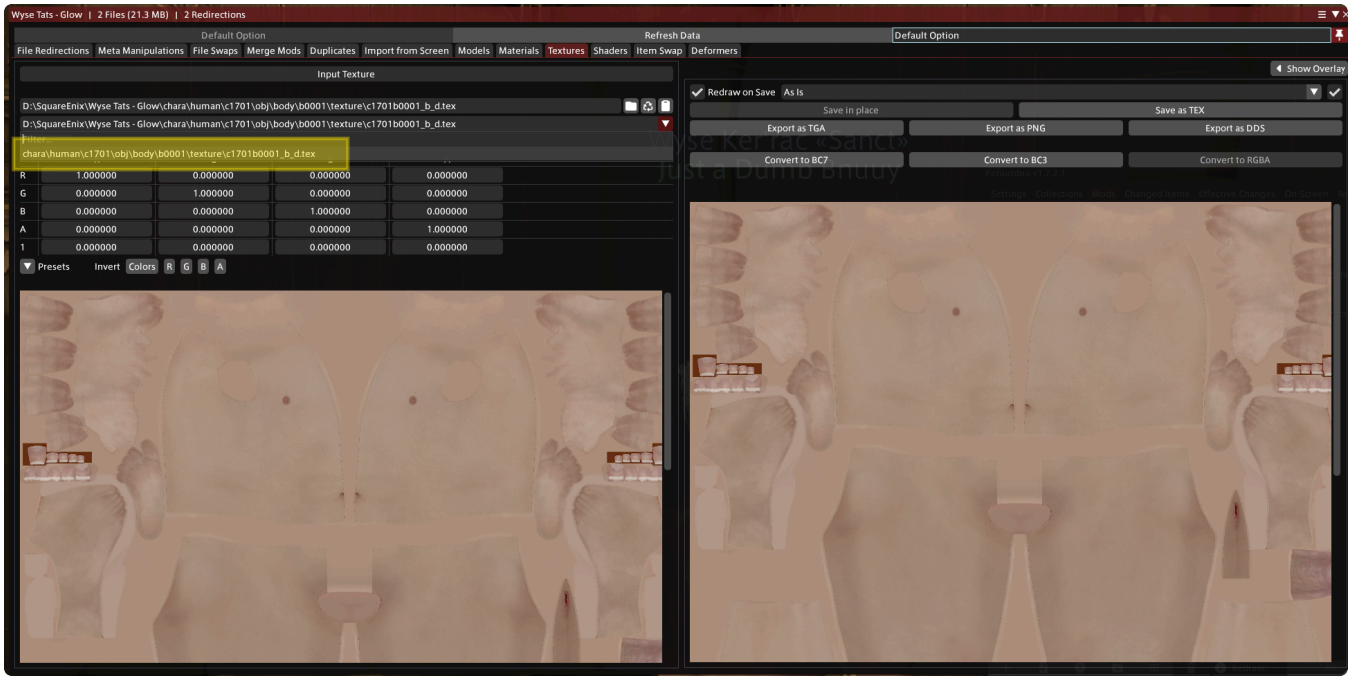
1. Back in **Import from Screen**, find the same material's **g_SamplerDiffuse** row (its color texture). The right-hand column shows it comes from your tattoo mod. Click its **import button**.



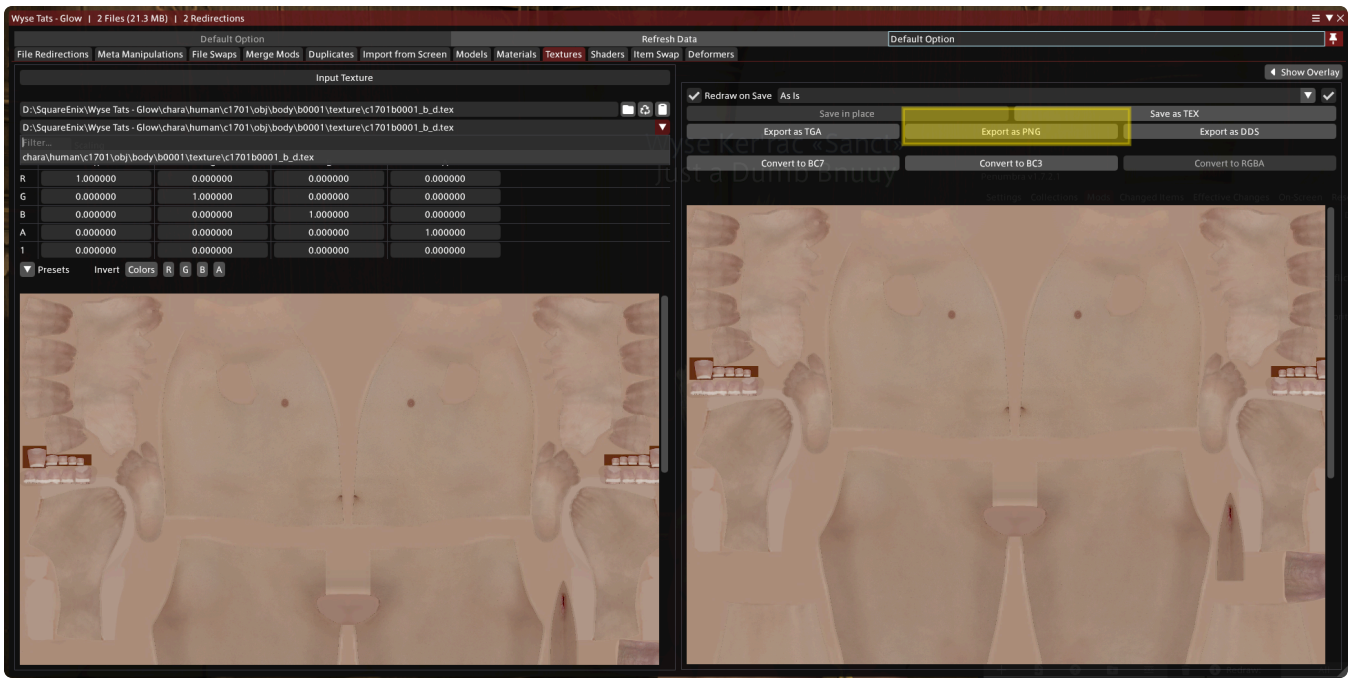
2. Open the **Textures** tab.



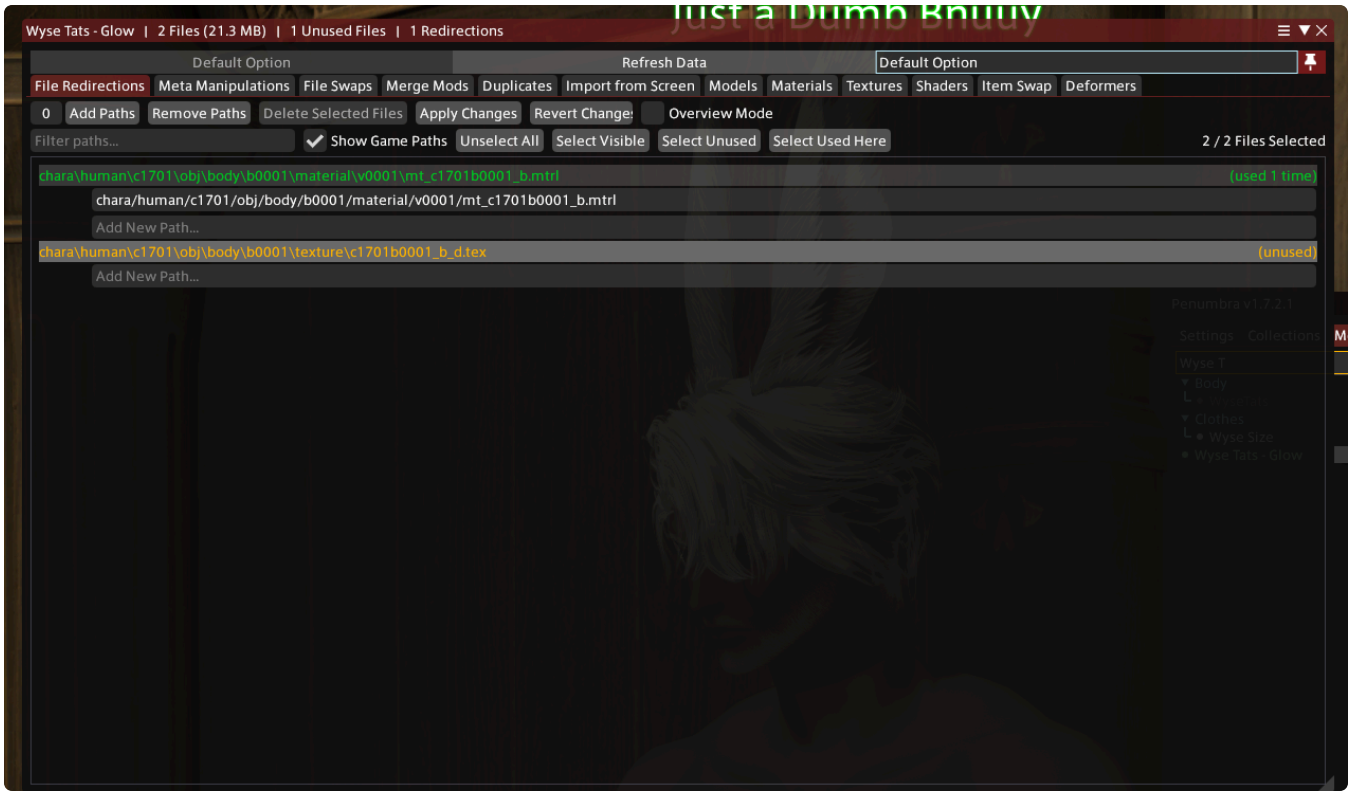
3. Open the dropdown under **Input Texture** and pick the color texture you just imported. The preview shows the body with the tattoo.



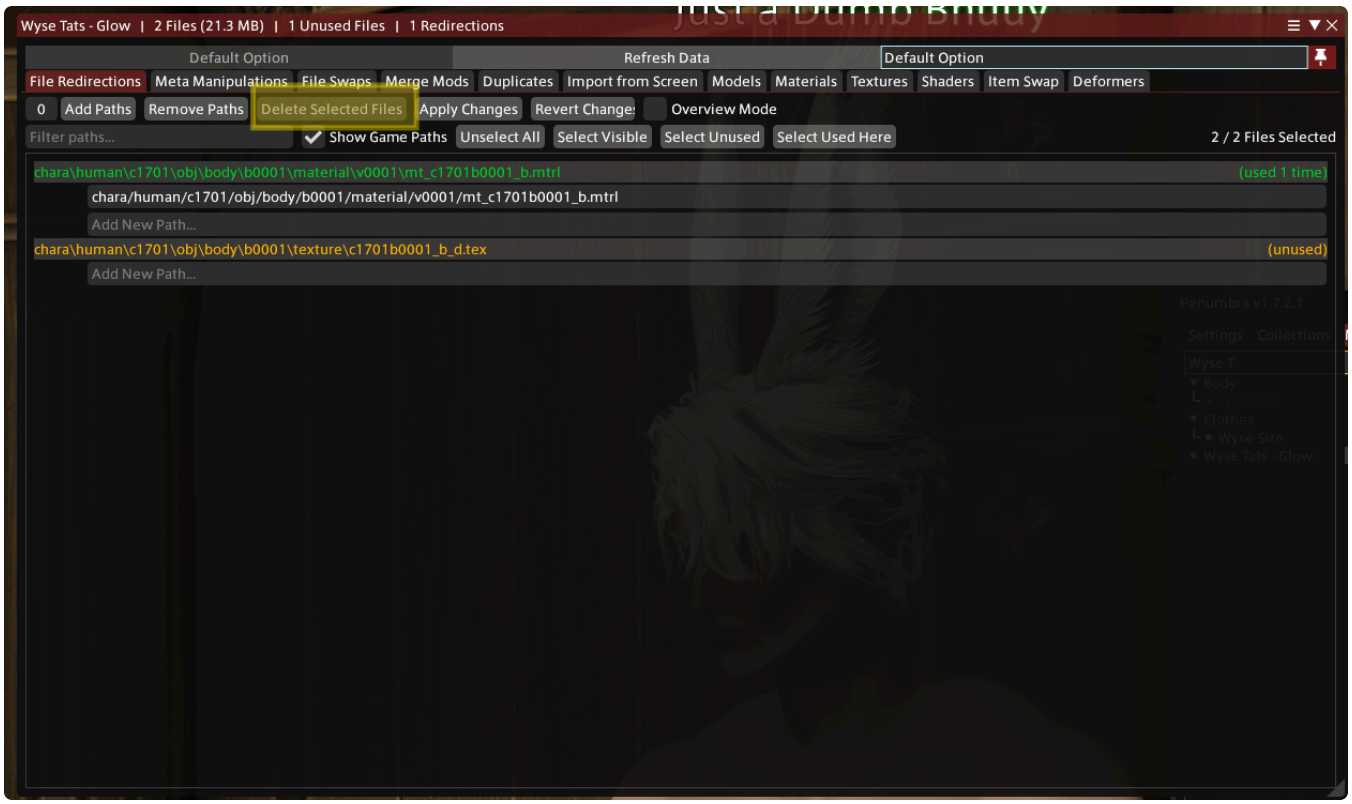
4. Click **Export as PNG** and save it somewhere handy. This is your layout reference.



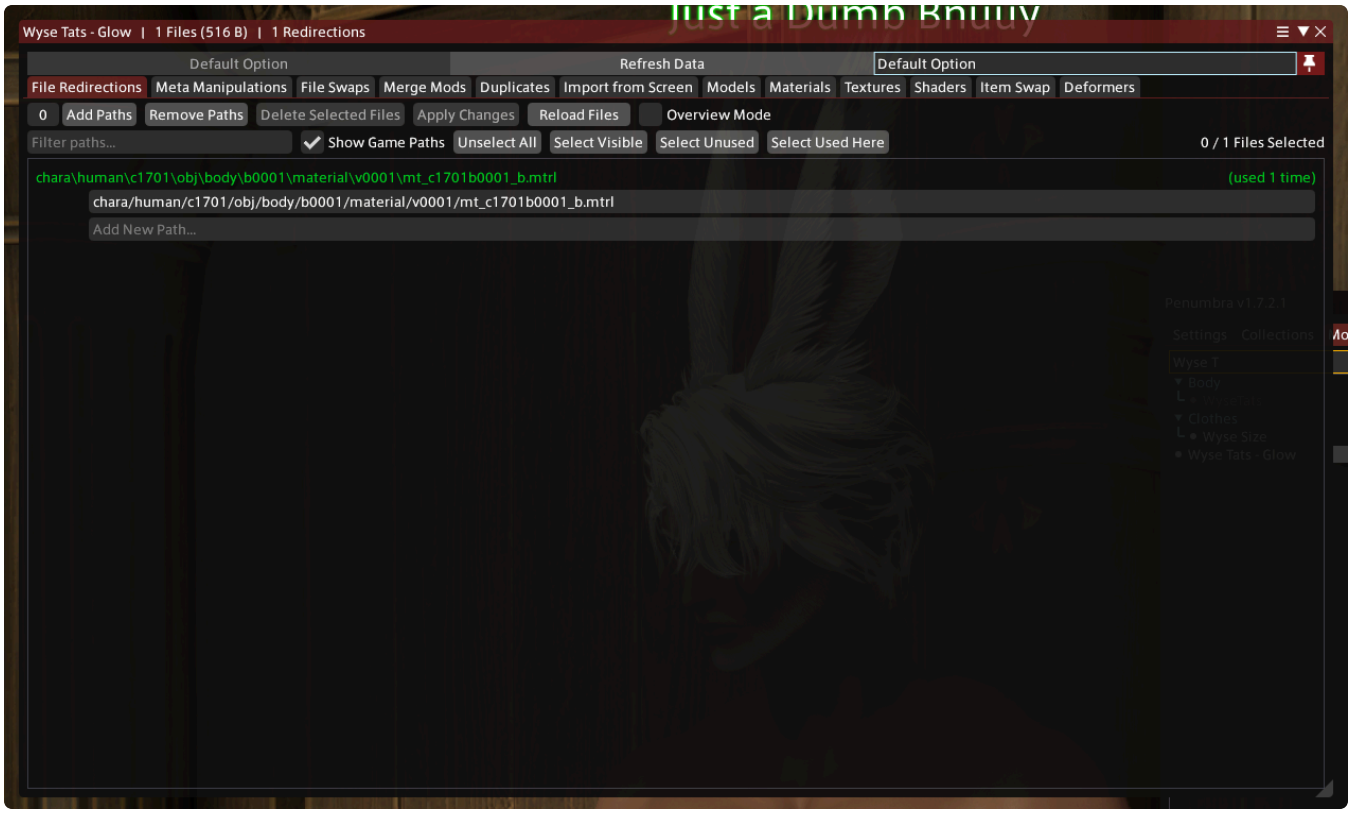
5. **Remove the color texture from the glow mod again. Don't skip this.** It was only needed for the reference. Left in, it overrides your tattoo mod's color texture, and if anything is ever saved over it, the skin itself changes color. Open **File Redirections**:
   - Clear the color texture's game path (empty its box). It now shows as **unused**.



Then click **Unselect All**, click the color texture (`..._d.tex`) so **only it** is selected, and click **Delete Selected Files**. Never select the material here.

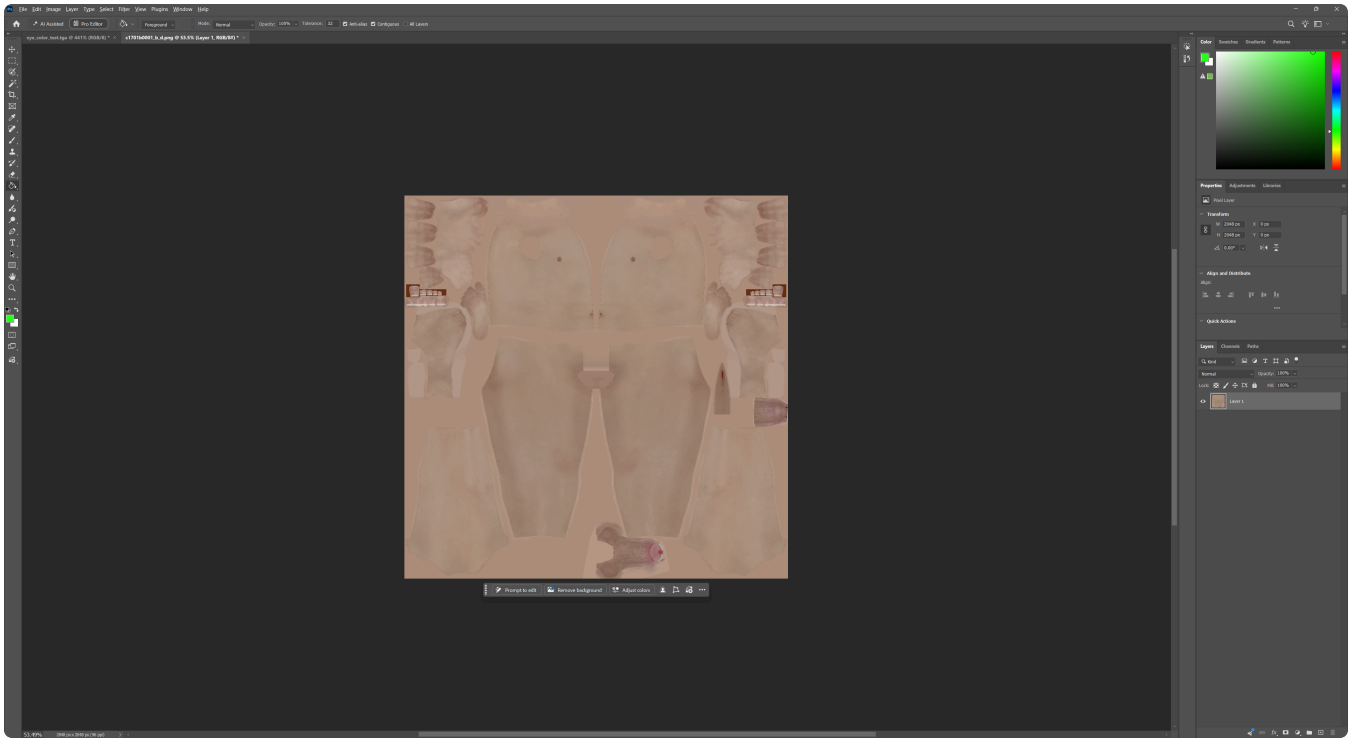


The mod is back to a single file: the material.

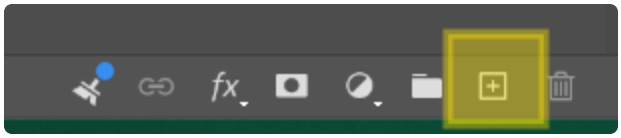


Now paint the glow. The screenshots use Photoshop; other editors work the same way with layers.

6. Open the reference PNG. It opens as a single layer.



7. Click **Create a new layer** at the bottom of the Layers panel.



8. A new, empty layer appears above the reference. Keep it selected.

Image — The new empty layer above the reference

9. On the new layer, paint the parts that should glow, in the colors they should glow. Here, blue on one side and green on the other. Paint over the tattoo's lines, or anywhere you like: the glow doesn't have to follow the ink.
    - Opacity is glow strength: soft or partly transparent strokes glow less.
    - Bold shapes read best from a distance; fine detail shows up close.

Image — The glow painted over the reference

10. **Hide the reference layer** by clicking its eye icon, so only the glow layer stays visible.

Image — The reference layer hidden in the Layers panel

The glow now sits alone on the transparent checkerboard. Everything shown as checkerboard won't glow.

Image — The glow layer alone on transparency

11. Choose **File > Export > Quick Export as PNG** and save the image, for example as `my_tattoo_glow.png`. Quick Export saves only what's visible and keeps the transparency.

Image — File > Export > Quick Export as PNG

Keep the image the same size as the reference (usually 2048×2048 or 4096×4096), so the glow lines up exactly with the body.

Step 3: Turn the PNG into a game texture

1. In the **Textures** tab, click the **folder icon** next to the input path.

Image — The folder icon for opening an image

2. Pick your glow PNG and open it. The preview shows the glow lines on a dark background (the transparent parts).

Image — Choosing the glow PNG

3. Open the output format dropdown at the top right (it starts as **As Is**)...

Image — The output format dropdown

4. ...and choose **BC7 (Complex Compression for RGBA)**. BC7 keeps the colors and the transparency at a quarter of the size of an uncompressed texture.

Image — BC7 in the format list

5. With BC7 selected, click **Save as TEX**.

Image — BC7 selected, Save as TEX

6. In the save dialog, go into your glow mod's folder, then `chara`, `human`, `c1701`, `obj`, `body`, `b0001` (the same folders as the material, but ending in **`texture`** instead of `material`). If there's no `texture` folder yet, click the **new folder** button, type `texture`, and click **Ok**.

Image — Creating the texture folder

7. Inside `texture`, **type a new name** into the **File Name** box, for example `WyseGlow`, and click **Ok**.

Image — Saving the glow as a new file in the texture folder

Always save the glow under a new name. Never click an existing file in this dialog: saving over the body's color texture (`..._d.tex`) turns your glow painting into the skin itself.

8. Open **File Redirections** and click **Reload Files**. The new texture appears, marked **unused**, because no game path points to it yet.

Image — The new glow texture, unused

9. In the **Add New Path...** box under it, type its game path: the folders and name in lowercase, with forward slashes. For the example: `chara/human/c1701/obj/body/b0001/texture/wyseglow.tex`

10. Click **Apply Changes**. Both files are now green and in use: the material and the glow layer.

Image — The material and the glow layer, both in use

The glow layer's game path is new: no vanilla file has it. That's fine, since it only needs to match what you enter in the material in the next step.

Step 4: Attach the glow layer to the material

1. Open the **Materials** tab and pick the material from the **Select .mtrl File...** dropdown at the top.

Image — Selecting the material in the Materials tab

2. Under **Textures and Samplers**, besides the material's usual textures (`g_SamplerDiffuse`, `g_SamplerMask`, `g_SamplerNormal`), there's a row named **g_SamplerLuminis**. Penumbra adds it automatically because the Luminis shader can use it, and fills it with a placeholder, `common/graphics/texture/dummy.tex`.

Image — The g_SamplerLuminis row with the dummy placeholder

3. Click into that row's path, replace `dummy.tex`'s path with your glow layer's game path from Step 3, and press **Enter**:
   `chara/human/c1701/obj/body/b0001/texture/wyseglow.tex`

Image — The g_SamplerLuminis row pointing at the glow layer

If there's **no g_SamplerLuminis row**, the material isn't using the Luminis shader. Check that the Luminis mod is enabled, with a higher priority than anything else that replaces `skin.shpk`.

There's also a **g_SamplerLuminis2** row, for an optional second glow layer (see Adding a second glow layer). Leave it on `dummy.tex` for a single layer.

Step 5: Turn on the glow

Still in the material editor, expand **Material Constants**, then **Luminis Glow**. Set:

Setting	Value	Why
Glow area	**Glow layer**	Glow where the glow layer is painted
Color source	**2**	Use the glow layer's own colors
Glow intensity	**1.5**	Brightness; 0 means no glow at all
Glow tint	**White**	Keeps your painted colors as they are
Glow mode	**Glow in darkness**	Or **Always on** while testing

Leave the rest at their defaults. Every setting has a hover description, and the Creator Guide's settings reference explains each one.

Skin materials start with **Glow intensity 0** and **Color source 1**, so nothing glows until you change them.

Click **Save to File** at the top of the material editor. Changes in the editor apply live, but only **Save to File** writes them into the mod. With **Redraw on Save** ticked, your character updates right away.

Image — The Luminis Glow settings for a glow layer, and Save to File

Step 6: Enable and test

1. Enable the glow mod and give it a **higher priority** than your body mod and tattoo mod (and than Luminis), so its material wins.
2. Type `/penumbra redraw self`. If the glow doesn't appear, restart the game once: textures that were already loaded can stay cached until then.
3. Find a dark spot (or set **Glow mode** to **Always on**) and look at your tattoo.

Image — The finished glow: the painted lines glowing on normal skin

4. Turn the glow mod **off** and redraw: the glow disappears, and the tattoo and skin look exactly as before.

Image — The same character with the glow mod turned off

If something looks wrong, the **Debug view** setting shows what Luminis sees. Set it back to **Off** when done.

- **Glow mask:** white where it would glow, black elsewhere. Only your painting should be white. A gray body means the glow layer isn't fully transparent: a reference or background layer was still visible when you exported.
- **Light gate:** white where the glow is currently on. Use it to tune **Dark start** and **Dark end**.

Image — Debug view: Glow mask, with only the painted lines in white

Other common fixes: if only part of the body reacts, the glow is on a material that covers only part of the mesh, so import the right one. If the skin itself changes color, check in the On-Screen tab that the material's color texture still comes from the body mod.

More things to check:

- **Settings show in Penumbra but change nothing:** the game is still running an older Luminis. Fully close and restart the game after installing or updating Luminis.
- **Red text next to Reload live preview, "The current material has not been found on your character":** your character isn't using this material. Another mod probably replaces it with a higher priority. In the **On-Screen** tab, the material row's right-hand column shows which mod wins. Raise your glow mod's priority above it.
- **Missing Files in the mod's title bar:** a file name and its game path don't match. Open the **Missing Files** tab, and fix the name or the path so they match exactly.
- **Glow area looks ignored:** check that the glow layer's path is in the **g_SamplerLuminis** row, not in **g_SamplerDiffuse** (that row is the item's own color texture, and must never change).

Variation: glowing eyes

Eyes are the quickest glow mod: a **natural glowing iris** needs no painting at all. Custom designs use a glow layer, just like the tattoo.

The eye material. In **Import from Screen**, the eye appears as its own entry, for example **Viera Male Face (Iris) 1**. The entry itself is the eye material (named like `mt_c1701f0001_iri_a.mtrl`), with its textures listed underneath. Click the import button on that entry's row (Step 1). Each race and face has its own eye material.

Image — The eye material in Import from Screen

A natural glowing iris (no painting). Open the eye material in the **Materials** tab and, under **Luminis Glow**, set **Glow intensity** to about 1.5 and **Glow mode** as you like. The defaults (**Glow area: Natural iris**, **Color source 1**) make only the iris glow, in the character's own eye color, while the pupil and whites stay normal. For a different color, set **Color source** to 0 and choose a **Glow tint**. Click **Save to File**. That's the whole mod.

Image — A natural glowing iris

A custom design (glow layer). Follow Steps 2 to 5 with these differences:

- **Step 2:** the layout reference is the eye's **g_SamplerMask** texture (`chara/common/texture/eye/eye01_mask.tex`). Import it, export it as PNG, and **remove the copy right away**: eye textures are shared by every character's eyes. The blue circle in it marks the iris.
- **Painting:** the eye mask is only 128×128, so first scale it up to **512×512** (Image > Image Size). On a new layer, select the iris with the **Elliptical Marquee** (hold Shift for a circle) and fill it, or use an **Angle** gradient with a spectrum preset for a rainbow. Delete a small circle in the middle to keep the pupil dark.
- **Step 3:** save the glow texture under a new name, for example `WyseEyeGlow`, in a `texture` folder next to the eye material's folder, and redirect that path.
- **Steps 4 and 5:** put the glow layer's path in the **g_SamplerLuminis** row, then set **Glow area: Glow layer** and **Color source: 2**.

Image — An eye glow layer: a green ring with a dark pupil

The painted design follows the iris, including as the pupil changes size.

Already using a "glow editable" eye mod? Those mods have their own always-on glow, set with **g_EmissiveColor**. To let Luminis control it instead (glow in darkness, animation), open the mod's eye material, set **Glow area: Emissive mask**, set **g_EmissiveColor** to black under **Further Constants**, and pick a **Glow tint**. Leave the eye's texture rows as they are, since the eye mod supplies them.

Variation: glowing hair

Hair usually needs **no painting at all**: only Step 1 (import the hair material, named like `mt_c1701h0004_hir_a.mtrl`), Step 5 and Step 6.

In Step 5, pick what glows with **Glow area**:

Glow area	What glows
Whole hair	Every strand
Highlights only	The highlight streaks
Main color only	Everything except the highlights
Luminis texture	A painted glow layer, as in Steps 2 to 4

Then choose the colors:

- **One color:** set **Color source** to 0 and pick a **Glow tint**.
- **Two colors**, for example cyan hair with red highlights: set **Color source** to 0, choose the main color as **Glow tint**, set **Separate highlight color** to **On**, and choose **Highlight glow color**.
- **The hair's own colors:** set **Color source** to 1.
- **Rainbow along the strands:** set **Color source** to 3, and adjust **Rainbow length** and **Rainbow shift**.

Image — The hair's Luminis Glow settings for two colors

Image — Two-tone glowing hair: cyan with red highlights

For painted hair glow, follow Steps 2 to 4 using one of the hair's textures (its normal or mask texture) as the layout reference, put the glow layer in the hair material's **g_SamplerLuminis** row, and set **Glow area** to **Luminis texture** and **Color source** to 2.

Variation: glowing gear, minions and mounts

Gear, minions, mounts and weapons work like tattoos, with two extra ways to choose what glows. They need the **Gear** and **Gear (newer)** parts of Luminis turned on.

The material. In **Import from Screen**, find the item (gear under its slot, a minion or mount as its own entry) and import its material row, as in Step 1. In the **Materials** tab, the **Shader** section shows `characterlegacy.shpk` or `character.shpk`: both work.

Under **Luminis Glow**, pick what glows with **Glow area**:

Glow area	What glows	Painting needed
Glow layer	Your painted design	Yes, as in Steps 2 to 4
Whole item	Everything using this material	No
Color table emissive	Each color table row, in its own Emissive Color	No

Color table emissive, step by step. Items are split into up to 16 color table rows (fabric, trim, a logo...), and each row can glow in its own color.

1. Find the rows: in the material's **Color Table**, hover over the **crosshair** button at the left of a row, and that row's part lights up on your character.
2. Give each row that should glow an **Emissive Color** (row A, B or both). Rows left black don't glow.
3. Under **Luminis Glow**, set **Glow area: Color table emissive**, **Color source: 2** (the rows' own colors) and **Glow intensity** around 2. Soft emissive colors need more, up to about 6.
4. Pick a **Glow mode**, and an **Animation** if you like, then click **Save to File**.

The item's own always-on row glow switches off automatically in this mode, so Luminis is fully in charge.

Image — Hovering a row's crosshair button in the Color Table highlights that row's part…

One row at a time (Luminis Rows). The **Luminis Rows** section has a dropdown for each of **Row 01** to **Row 16**: **Animated** follows the Animation, **Static** glows steadily in the row's Emissive Color, and **Off** never glows. For example, with Chroma animation, a hoodie's fabric row on **Static** (Row 01) stays one steady color, while its logo row (Row 08) on **Animated** cycles through the rainbow.

Set rows the item doesn't use to **Off**, so nothing unexpected lights up.

Image — Luminis Rows: Row 01 on Static, Row 08 on Animated, the unused rows Off

Image — In-game: the hoodie fabric steady, the logo on the back cycling through the rai…

A glow on your minion or mount uses the same collection as your character, so it shows whenever your character's mods do.

Animating the glow

Any glow can blink, alternate colors or cycle through the rainbow. In the material's **Luminis Animation** section:

1. Pick an **Animation**:
   - **Blink:** on and off.
   - **Pulse:** like Blink, but breathes between **Pulse minimum** and **Pulse maximum** instead of going dark. 0.15 means 15% of full brightness.
   - **Two-color:** fades between **Glow tint** and **Color 2**.
   - **Color steps:** evenly spaced rainbow colors.
   - **Chroma:** a smooth rainbow; the whole glow changes color together.
   - **Custom colors: Glow tint** and **Colors 2 to 4**.

- **Hue cycle:** every color rotates around the color wheel at once, so a painted rainbow gradient flows. Use it with **Color source 2**.

2. Set **Speed**: 0.005 is barely noticeable, 0.05 is a slow drift, 0.2 is a gentle cycle, and larger values strobe.
3. Set **Softness**: 0 switches hard, 1 fades smoothly. For Color steps and Custom colors, **Steps** sets how many colors.
4. Click **Save to File**.

Animated colors follow the brightness of your painting: dark parts stay dark, while bright parts of any color take the animation colors at full strength. To keep a painted gradient's colors and make them flow, use **Hue cycle** rather than **Chroma**.

Image — A painted rainbow tattoo with Hue cycle, at three moments: every color moves ar…

Image — The Luminis Animation section set to Custom colors

Adding a second glow layer

A second glow layer has its own settings, so one design can stay **always on** while another only glows **in darkness**, or animates on its own.

1. Paint the second design as its own PNG, and turn it into a game texture under a new name (Step 3), for example `WyseGlow2`.
2. In the **Materials** tab, put its path in the **g_SamplerLuminis2** row, the second `dummy.tex` row under Textures and Samplers.
3. In the **Luminis Layer 2** section, set **Layer 2 intensity** above 0, plus its **glow mode**, **color source** and **animation**. Layer 2 color source 0 gives pure animation colors.
4. Click **Save to File**.

Where both layers glow on the same spot, their glow adds up, so paint them side by side for separate effects.

Image — Textures and Samplers with both glow layer rows filled in

Image — The Luminis Layer 2 section, with its own intensity and animation

Image — Both layers glowing in-game: each with its own color and animation

Updating your tattoo later

The glow lives in its own mod, and its settings live in the material, so a new tattoo design only needs a new glow layer.

A new design on the same body:

1. Paint the new glow (Step 2), using a fresh layout reference from the new tattoo.
2. Convert it (Step 3), and save it under the **same name** as your old glow layer, in the same `texture` folder of your glow mod. Overwriting is fine here, because it's your own glow file. Never overwrite the body's `..._d.tex`.
3. In **File Redirections**, click **Reload Files**, then redraw.

The material and all its Luminis settings stay exactly as they were. To keep the old design too, save the new one under a new name instead, give it a game path, and swap it into the **g_SamplerLuminis** row; switching back is then one path change.

The body mod itself was updated (for example a new TBSE version with a new material): import the new material into your glow mod (Step 1), then redo Steps 4 and 5. A new material starts with fresh settings, so screenshot your **Luminis Glow**, **Luminis Animation** and **Luminis Layer 2** sections first, and enter the same values again.

Sharing your glow mod

1. Select the glow mod and open its **Edit Mod** tab. Fill in the **Name**, **Author**, **Version** and **Description**.
2. In the description, tell users: *Requires Luminis 2.0 or later (with its Gear part, for gear, minions and mounts). Load order: body mod, then tattoo mod, then Luminis, then this glow mod. Glows in darkness by default; fine-tune it in Penumbra's material editor under Luminis Glow.*
3. Click **Export Mod**. Penumbra creates a `.pmp` you can upload.

Image — The Edit Mod tab, with Edit Description and Export Mod

A glow mod covers only the body mod and options its material came from. To support both TBSE and Bibo, or several skin types, import each one's material into an **option group** in the same mod.